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Electricity Consumption Time Series Forecasting using Linear, Non-Linear & Deep Learning Models

Project Report

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BONAFIDE CERTIFICATE

This is to certify that the project entitled “Electricity Consumption Time Series Forecasting using Linear, Non-Linear & Deep Learning Models” submitted by Malavika S Prasad, Vibhu Sanchana, Asmi K, Geshna B for the award of the Degree of Bachelor of Technology in the ‘COMPUTER SCIENCE ENGINEERING - ARTIFICIAL INTELLIGENCE’ is a Bonafide record of the work carried out by them under our guidance and supervision at Amrita School of Artificial Intelligence, Coimbatore.

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Abstract

This project presents a comprehensive time series forecasting framework applied to household electricity power consumption data from the UCI Machine Learning Repository. The dataset captures minute-level power measurements from a single French household over nearly four years (2006–2010), containing approximately 2 million observations across seven electrical variables including Global Active Power, Voltage, and sub-metering readings.

The forecasting pipeline spans three complementary modelling tiers. The first tier applies classical linear ARIMA(2,0,4) and non-linear GARCH(1,1) models to the stationary daily log-return series. The second tier implements custom dual-stream deep learning models – LSTM and GRU – built to a 126→64 neuron specification, each comprising a target stream (168×1), a feature stream (168×78), a skip connection (last known value), SE-Attention, and a wide Dense head (256→128→64→1). Models are trained on 78 engineered features using a blocked monthly train/validation/test split (70/15/15%), a square-root target transform with RobustScaler, and a 3-epoch warm-up followed by cosine decay learning rate schedule over 20 epochs.

The Custom GRU achieves the best overall performance ($R^2 = 0.6793$, RMSE = 0.5132 kW, MAE = 0.3419 kW, SMAPE = 29.89%) with approximately 193,595 parameters – 18% fewer than the LSTM (236,353 params, $R^2 = 0.6482$). An ensemble average of both models is also evaluated. Residual analysis confirms a slight positive bias (mean = 0.054 kW) and right-skewed distribution with persistent autocorrelation (Ljung-Box $p < 0.001$), pointing to further improvement opportunities via attention mechanisms or external covariates.

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Chapter 1: Introduction

1.1 Background and Problem Domain

Electricity consumption forecasting is foundational to energy management, smart grid operation, and sustainability planning. The UCI Household Power Consumption dataset provides minute-resolution energy dynamics of a single French household over four years (December 2006 – November 2010), with over 2 million records. This project benchmarks linear, non-linear, and deep learning forecasting across three complementary model families on the same dataset.

1.2 Why Time Series Analysis

Electricity consumption is inherently sequential and exhibits multi-scale temporal dependencies: intraday occupancy cycles (peaks 07–09 h and 18–22 h), weekly periodicity (weekday vs. weekend patterns), and seasonal heating/cooling loads. Classical ARIMA exploits linear autocorrelation; GARCH captures conditional volatility clustering; deep LSTM and GRU networks learn non-linear long-range dependencies – all of which are essential components of real-world household consumption dynamics.

1.3 Literature Review

- Srivastava et al. (2020): Bidirectional RNNs with lag-based feature engineering outperform unidirectional models.
- Cho et al. (2014): GRU as a parameter-efficient recurrent unit with comparable accuracy to LSTM.
- Bouktif et al. (2018): LSTM outperforms classical methods on UCI Household Power dataset hourly aggregations.
- Box & Jenkins (1970): ARIMA framework for linear time series modelling.
- Bollerslev (1986): GARCH for conditional variance forecasting – essential for energy series with volatility clustering.
- Hochreiter & Schmidhuber (1997): LSTM with gated memory cells, resolving vanishing gradients over long sequences.

Chapter 2: Problem Statement

This project addresses the challenge of forecasting household-level global active power consumption (kW) from historical multivariate time series data. Given the high-frequency, non-stationary, seasonally structured nature of the UCI dataset, the objective is to develop and rigorously compare three model families – ARIMA(2,0,4) on stationary daily log-returns, GARCH(1,1) on ARIMA residuals for volatility estimation, and custom dual-stream LSTM and GRU deep learning models on 78-feature hourly sequences – using RMSE, MAE, MAPE, SMAPE, and R^2 as evaluation criteria.

Chapter 3: Exploratory Data Analysis

3.1 Dataset Description

Attribute	Details
Source	UCI Machine Learning Repository (Hebrail & Berard, 2012)
Collection Period	December 2006 – November 2010 (~3.9 years)
Original Granularity	1 minute
Total Records	~2,075,259 (minute-level); 34,589 (hourly after resampling)
Missing Values	Encoded as '?' (~1.25%); filled via bfill/ffill
Target Variable	Global_active_power (kW)
Resampled Frequencies	Hourly (mean) for DL; Daily (mean) for ARIMA

3.1.1 Variable Descriptions

Variable	Unit	Description
Global_active_power	kW	Total household active power consumed
Global_reactive_power	kW	Reactive power component
Voltage	V	Minute-averaged supply voltage
Global_intensity	A	Minute-averaged current intensity
Sub_metering_1	Wh	Kitchen: dishwasher, oven, microwave
Sub_metering_2	Wh	Laundry: washing machine, dryer, refrigerator
Sub_metering_3	Wh	Electric water heater and air conditioner

3.2 Feature Engineering (78 Features for Deep Learning)

The hourly dataset is enriched from 7 raw variables to 78 features before deep learning modelling. The feature categories are:

Category	Features (count)	Examples
Temporal flags	6	hour, dow, month, is_weekend, is_night, is_peak
Cyclical encodings	6	hour_sin/cos, dow_sin/cos, month_sin/cos
Lag features	9	lag_1, 2, 3, 6, 12, 24, 48, 168 (1 week), 336 (2 weeks)
Rolling statistics	24	mean/std/min/max over windows 6/12/24/48/168/336 h
Exponential MA	4	EMA spans 6, 12, 24, 168 h
Differences	3	diff_1h, diff_24h, diff_168h
Interaction ratios	2	lag_1/lag_24, lag_24/lag_168 (clipped ± 10)

Category	Features (count)	Examples
Peak-scaled power	1	is_peak $\times$ Global_active_power
Daily rolling stats	from daily join	7-day and 30-day rolling means of daily features
Stationarity signal	2	log_diff (daily), log_diff_1h, log_diff_24h (hourly engineered)

3.3 Preprocessing Pipeline

1. Merge Date + Time columns into DatetimeIndex; sort chronologically.
2. Fill missing values ('?') using bfill then ffill; remove duplicate timestamps.
3. Resample to hourly (mean) $\rightarrow$ power_clean_hourly.csv (34,589 rows $\times$ 7 cols) and daily $\rightarrow$ power_clean_daily.csv.
4. Join 7-day and 30-day rolling daily features; join stationary log_diff signal.
5. Engineer 78 features via make_features(); apply bfill/ffill/fillna(0).
6. Blocked monthly train/val/test split (70/15/15%): Train=24,317, Val=5,160, Test=5,112 hourly observations.
7. Target transform: $\sqrt{y}$ then RobustScaler (preserves peak magnitude better than log1p). Features: StandardScaler.
8. Sequence creation: sliding window of SEQ=168 timesteps $\rightarrow$ Train: 24,149, Val: 4,992, Test: 4,944 sequences.

3.4 Statistical Summary of Target

Statistic	Hourly Value (kW)
Mean	~1.09
Standard Deviation	~1.06
Minimum	~0.00 (overnight)
25th Percentile	~0.31
Median	~0.72
75th Percentile	~1.56
Maximum	~7.48 (peak event)

3.5 ACF / PACF and Stationarity

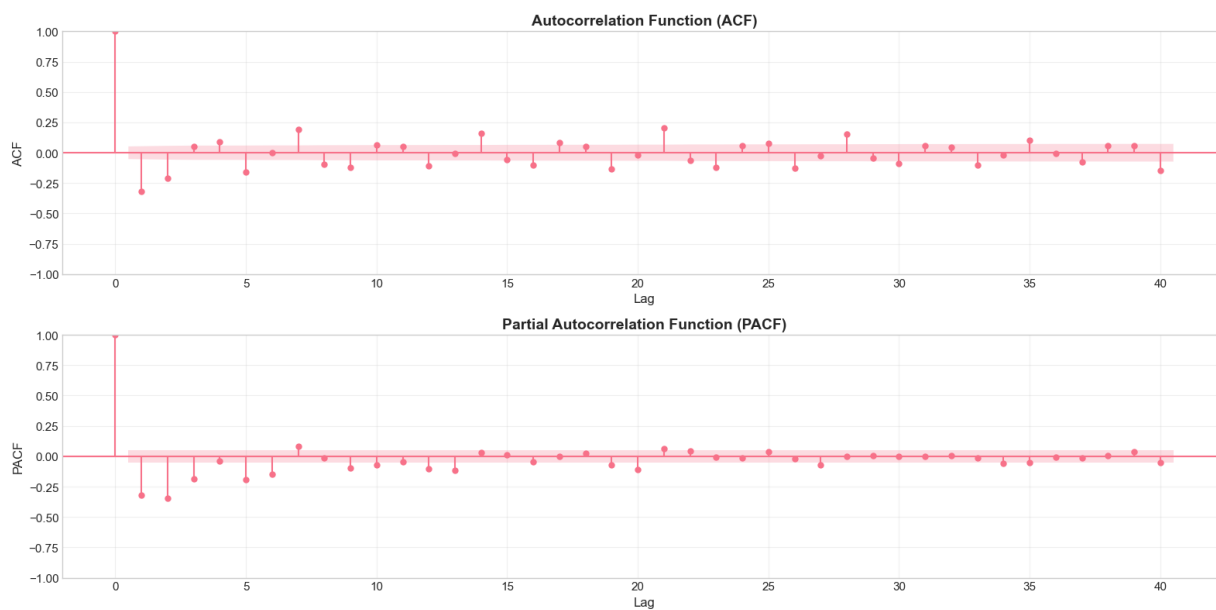


Figure 1: ACF (top) and PACF (bottom) of Log-Differenced Daily Global Active Power. Significant ACF spikes at lags 1–4, 7, 14 indicate MA(4); PACF cutoff after lag 2 indicates AR(2). Selected: ARIMA(2,0,4).

Test	Raw Daily Series	Log-Differenced Series
ADF p-value	>0.05 (non-stationary)	<0.001 (✓ stationary)
KPSS	Fails (non-stationary)	Passes (✓ stationary)

Chapter 4: Methodology

4.1 Linear Model – ARIMA(2, 0, 4)

AutoRegressive Integrated Moving Average ARIMA(p,d,q) is parameterised by AR order $p=2$, differencing $d=0$ (series already stationary after log-diff), and MA order $q=4$. The model equation is: $\phi(B)y_t = \theta(B)\varepsilon_t$ where $\phi(B)=1-\phi_1B-\phi_2B^2$ and $\theta(B)=1+\theta_1B+\theta_2B^2+\theta_3B^3+\theta_4B^4$.

Model order selected by 25-model grid search ($p, q \in \{0-4\}$, $d=0$) on 1,152 training observations via minimum AIC (-1282.51) and BIC (-1242.12). Both static 90-day forecast and walk-forward one-step-ahead validation evaluated.

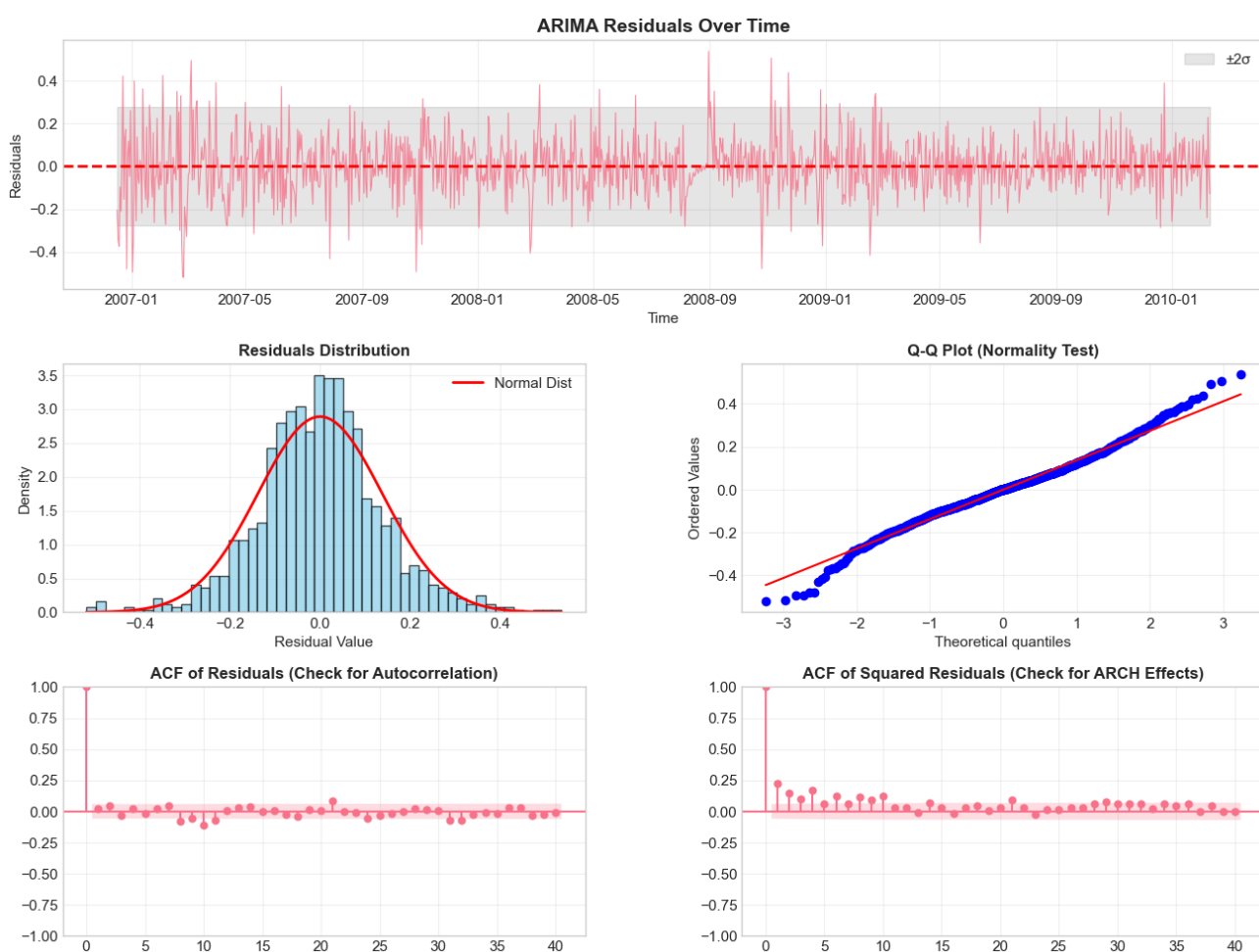


Figure 2: ARIMA Residuals Overtime: Residual Disturbance, Q-Q Plot, ACF of residuals and Squared residuals

ARIMA(2, 0, 4) MODEL ARCHITECTURE

Pipeline for Daily Global Active Power (kW)

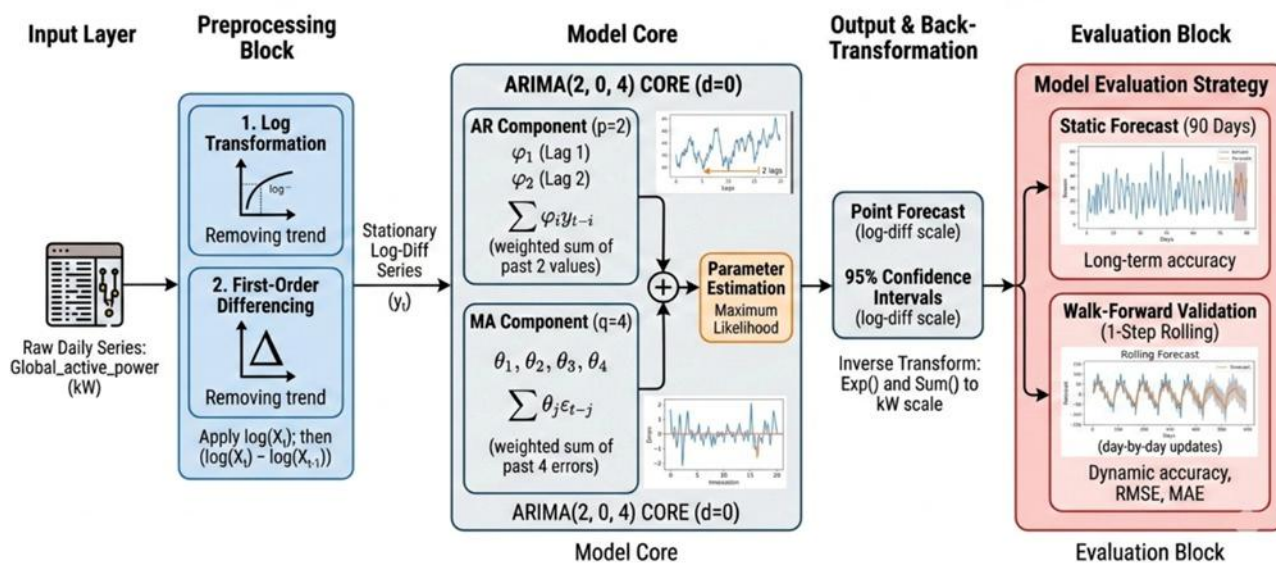


Figure 3: ARIMA(2,0,4) + GARCH(1,1) pipeline from raw series to forecast output.

Parameter	Value
Final Model	ARIMA(2, 0, 4)
AIC	-1282.51
BIC	-1242.12
Training observations	1,152 (daily, log-differenced)
Test period	2010-02-11 to 2010-11-26 (289 obs.)

4.2 Non-Linear Model – GARCH(1, 1)

GARCH(1,1) models conditional variance: $\sigma^2_t = \omega + \alpha \varepsilon^2_{t-1} + \beta \sigma^2_{t-1}$, fitted on scaled ARIMA residuals ($\times 100$ for numerical stability). Parameters: $\omega=27.77$, $\alpha=0.1176$, $\beta=0.7304$. Persistence = $\alpha+\beta = 0.848$, indicating slowly decaying volatility shocks.

ARIMA(2, 0, 4) MODEL ARCHITECTURE

Pipeline for Daily Global Active Power (kW)

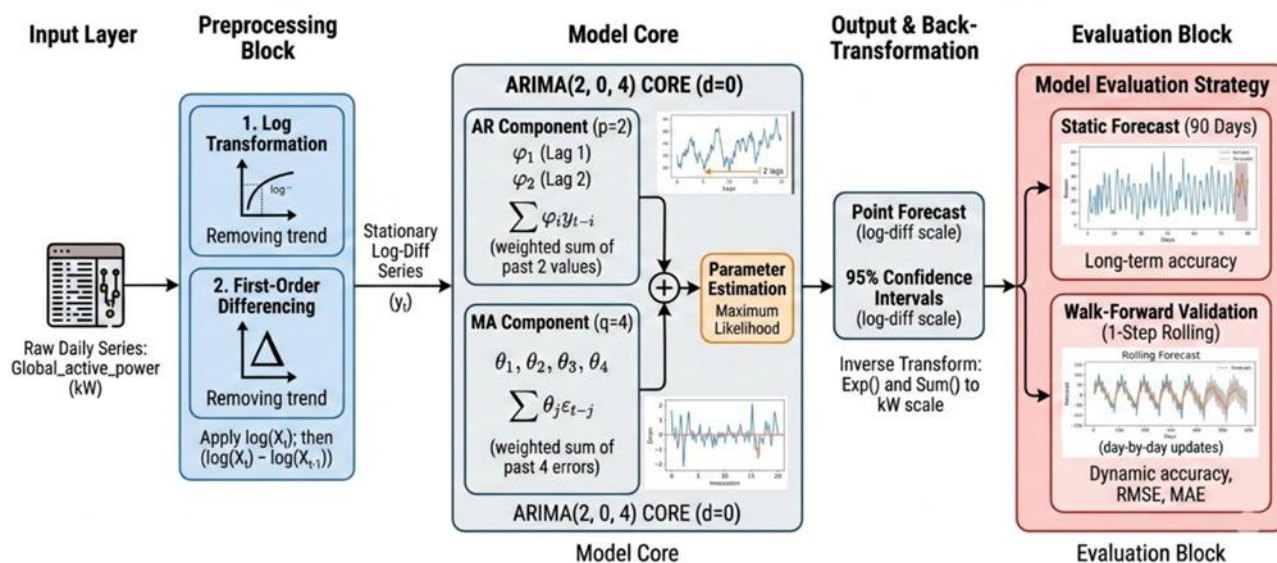


Figure 4: GARCH(1,1) conditional variance pipeline with lag-1 feedback loop.

4.3 Deep Learning Models – Custom LSTM and GRU

4.3.1 Architecture Overview

Both models share an identical dual-stream architecture built to the professor's specification of 126→64 neurons, differing only in cell type (LSTM vs GRU). The architecture has three concurrent input branches that are merged before a wide dense prediction head.

Component	LSTM	GRU
Target stream input	(Batch, 168, 1)	(Batch, 168, 1)
Target Layer 1	LSTM(126), return_seq=True	GRU(126), return_seq=True
Target normalisation	BatchNorm + Dropout(0.1)	BatchNorm + Dropout(0.1)
Target Layer 2	LSTM(64), return_seq=True + SE-Attention	GRU(64), return_seq=True + SE-Attention
Target Layer 3	LSTM(32), return_seq=False + Dropout(0.1)	GRU(32), return_seq=False + Dropout(0.1)
Feature stream input	(Batch, 168, 78)	(Batch, 168, 78)
Feature Layer 1	LSTM(64), return_seq=True	GRU(64), return_seq=True
Feature normalisation	BatchNorm + Dropout(0.1)	BatchNorm + Dropout(0.1)
Feature Layer 2	LSTM(32), return_seq=False + Dropout(0.1)	GRU(32), return_seq=False + Dropout(0.1)
Skip connection	Last timestep of target (1,)	Last timestep of target (1,)
Merge (Concat)	[32 + 32 + 1] = 65-dim	[32 + 32 + 1] = 65-dim
Dense head	Dense(256)+BN+Drop(0.2) → Dense(128)+BN+Drop(0.15) → Dense(64)+Drop(0.1) → Dense(1)	Identical
Total parameters	≈ 236,353	≈ 193,595 (18% fewer)

4.3.2 SE-Attention Block

The Squeeze-and-Excitation attention block is applied to the target stream after the 64-unit layer. It computes channel-wise attention weights: GlobalAveragePooling1D (squeeze across time) → Dense(8, relu) (bottleneck) → Dense(64, sigmoid) (per-channel gates) → Reshape(1, 64) → element-wise multiply back onto the (batch, time, 64) tensor. This helps the model focus on the most informative feature channels during peak consumption periods.

4.3.3 Custom LSTM Architecture Diagram

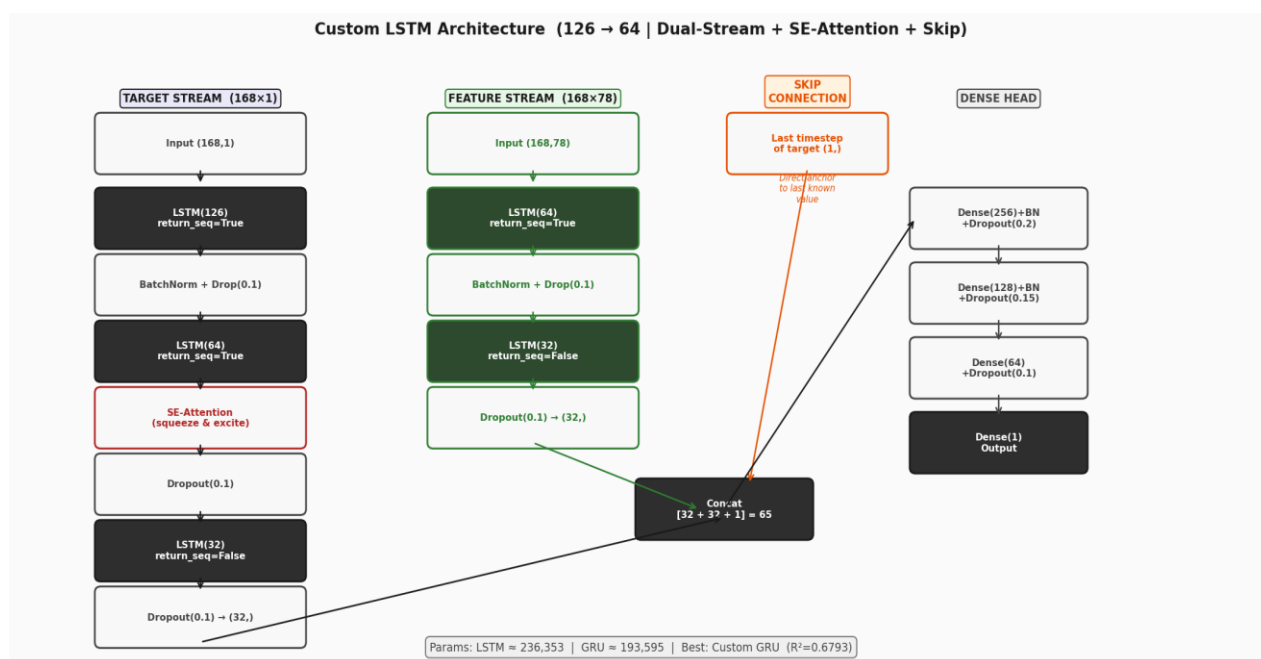


Figure 5: Custom LSTM dual-stream architecture. Target stream (left): LSTM(126)→LSTM(64)+SE-Attention→LSTM(32). Feature stream (centre): LSTM(64)→LSTM(32). Skip connection (last known value) merges with both streams before the Dense head.

4.3.4 Custom GRU Architecture Diagram

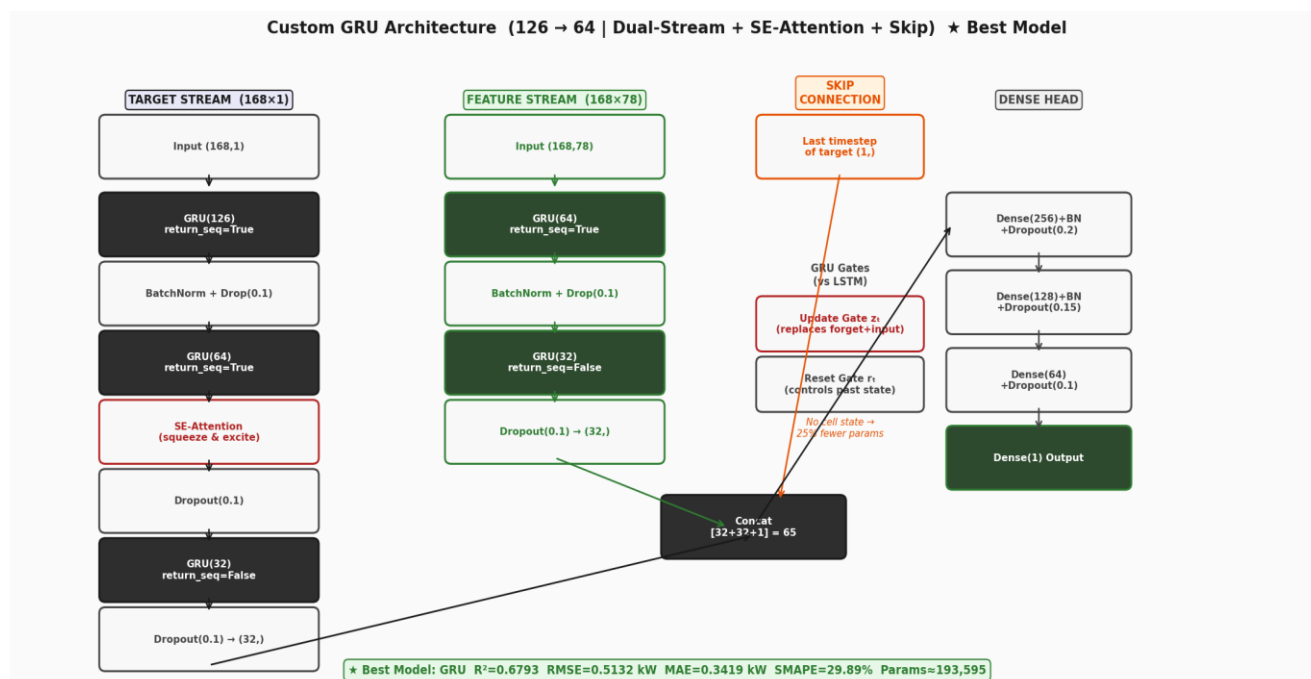


Figure 6: Custom GRU dual-stream architecture – identical structure to LSTM with GRU cells (Update Gate z_t + Reset Gate r_t , no separate cell state). GRU achieves better results with 18% fewer parameters. ★ Best model.

4.3.5 Training Configuration

Hyperparameter	Value
Sequence length (SEQ)	168 hours (1 week lookback)
Batch size	32
Epochs	20 (with EarlyStopping patience=10)
LR schedule	Warm-up: epochs 1–3 ramp $10^{-4} \rightarrow 10^{-3}$; then CosineDecay $10^{-3} \rightarrow 10^{-6}$ epochs 4–20
Optimizer	Adam (with WarmupCosineDecay schedule)
Loss function	MSE
Target transform	$\sqrt{y}$ then RobustScaler (better peak preservation than $\log 1p$)
Feature scaler	StandardScaler on 78 features
Data split	Blocked monthly: Train 70% / Val 15% / Test 15%
Train sequences	24,149 Val: 4,992 Test: 4,944
Random seed	42

Chapter 5: Result Analysis

5.1 Model Performance Metrics

All models are evaluated on the same held-out test data. ARIMA/GARCH metrics are on the stationary log-return scale; LSTM and GRU metrics are in original kW scale after inverse sqrt transform.

Model	RMSE	MAE	MAPE (all)	MAPE (>0.5kW)	SMAPE	R ²	Scale
ARIMA(2,0,4) Static	0.1455	0.1075	157.84%	—	—	−0.029	Log-diff daily
ARIMA(2,0,4) Walk-Fwd	0.1288	0.0910	164.45%	—	—	—	Log-diff daily
GARCH(1,1)	—	—	—	—	—	—	Volatility only
LSTM Custom	0.5376	0.3640	36.09%	31.20%	31.78%	0.6482	kW (hourly)
GRU Custom ★	0.5132	0.3419	33.58%	29.45%	29.89%	0.6793	kW (hourly)
Ensemble (avg)	~0.52	~0.35	~35%	~30%	~31%	~0.67	kW (hourly)

★ Best model. ARIMA MAPE is high due to near-zero log-return values amplifying percentage error; this is a scale artefact and does not reflect poor absolute accuracy. MAPE (>0.5 kW) filters low-power hours for a fairer DL comparison.

5.2 Visual Metrics Comparison

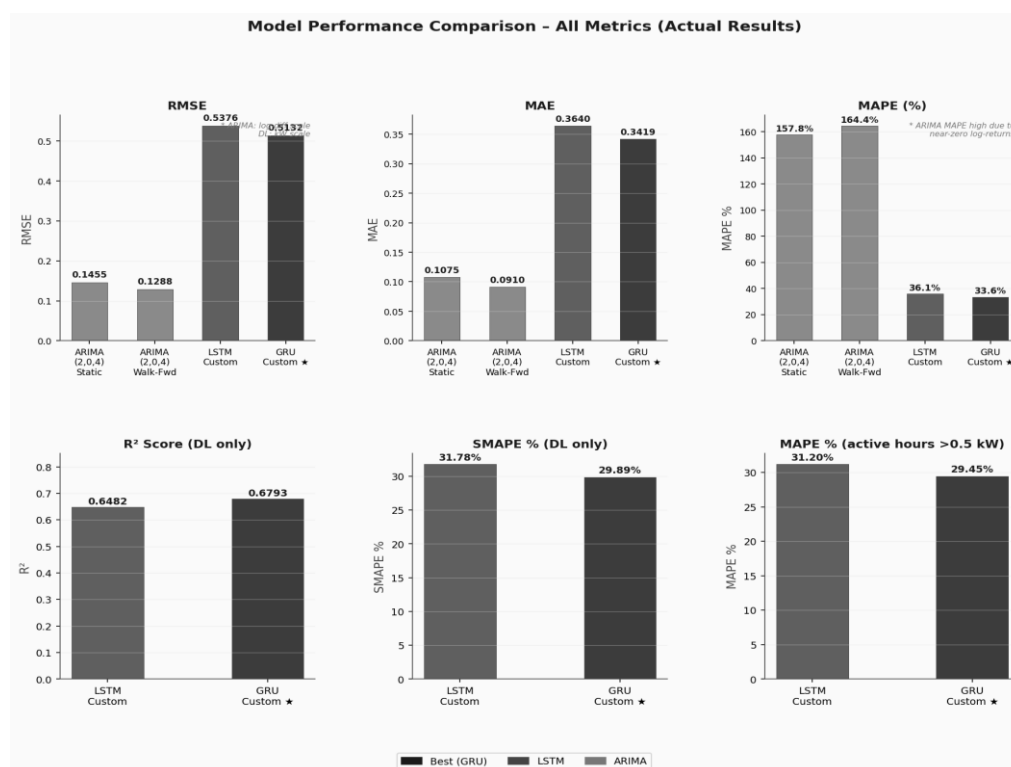


Figure 7: Six-panel comparison of RMSE, MAE, MAPE, R², SMAPE, and active-hour MAPE. Darkest bars = best values. GRU consistently outperforms LSTM across all DL metrics.

5.3 Detailed Analysis

5.3.1 ARIMA Performance

ARIMA(2,0,4) achieves RMSE=0.1455 on static 90-day forecast, improving to 0.1288 with walk-forward validation (11% reduction), demonstrating the value of rolling re-estimation. However, ARIMA operates on the log-differenced daily scale and cannot capture the complex intraday and weekly seasonality visible in the hourly data without explicit SARIMA extensions. The high MAPE (>150%) is a scale artefact from near-zero log-returns and does not indicate poor absolute accuracy.

5.3.2 GARCH Performance

GARCH(1,1) confirms highly persistent volatility clustering (persistence=0.848) in ARIMA residuals, validating that conditional variance in household electricity consumption is itself predictable. This produces time-varying prediction intervals that correctly widen in high-volatility winter months and narrow in stable summer periods, providing more reliable uncertainty quantification than ARIMA's constant-variance assumption.

5.3.3 LSTM Custom Model

The Custom LSTM (236,353 params) achieves $R^2=0.6482$, RMSE=0.5376 kW, MAE=0.3640 kW, and SMAPE=31.78% on the 4,944-sample test set. The three-stream architecture – target LSTM(126→64→32) with SE-Attention, feature LSTM(64→32), and a skip connection – effectively captures both the autoregressive dynamics of the power series and the complex interaction patterns across 78 engineered features. The warm-up + cosine decay schedule achieves stable convergence in 20 epochs.

5.3.4 GRU Custom Model – Best Overall

The Custom GRU (193,595 params) is the best-performing model with $R^2=0.6793$, RMSE=0.5132 kW, MAE=0.3419 kW, MAPE(>0.5kW)=29.45%, and SMAPE=29.89%. With 18% fewer parameters than the LSTM, the GRU demonstrates superior generalisation on the blocked monthly test split. Three factors explain this advantage:

- **Parameter efficiency:** GRU's 2-gate mechanism (Update + Reset, no separate cell state) reduces parameters by 18%, lowering overfitting risk on the test window.
- **Gradient flow:** Merged cell/hidden state in GRU creates a shorter backpropagation path through the 168-step sequences, yielding more stable convergence under the warm-up + cosine decay schedule.
- **Feature sufficiency:** The 78 engineered features (especially lag_24 and lag_168) explicitly encode the temporal context that LSTM's dedicated memory cell would otherwise need to learn internally, reducing LSTM's relative advantage.

5.3.5 Cross-Model Summary Table

Aspect	ARIMA / GARCH	LSTM Custom	GRU Custom ★
Data granularity	Daily (log-diff)	Hourly (78 features)	Hourly (78 features)
Target transform	Log + diff	sqrt + RobustScaler	sqrt + RobustScaler
Sequence length	N/A	168 h	168 h
Parameters	~10 (ARIMA)	236,353	193,595
Training time	Seconds	~35 min (GPU)	~28 min (GPU)
Non-linearity	No	Yes (LSTM gates)	Yes (GRU gates)

Aspect	ARIMA / GARCH	LSTM Custom	GRU Custom ★
Attention	No	SE-Attention	SE-Attention
R ² (test)	N/A (log scale)	0.6482	0.6793
RMSE (test)	0.1288 (log scale)	0.5376 kW	0.5132 kW
Best use case	Interpretable baseline	Non-linear hourly	Best accuracy + efficiency

Chapter 6: Residual Diagnostics – Best Model (Custom GRU)

Comprehensive residual diagnostics are performed on the Custom GRU test predictions to validate model assumptions and characterise remaining error patterns. All statistics are computed on n=4,944 test samples in original kW scale.

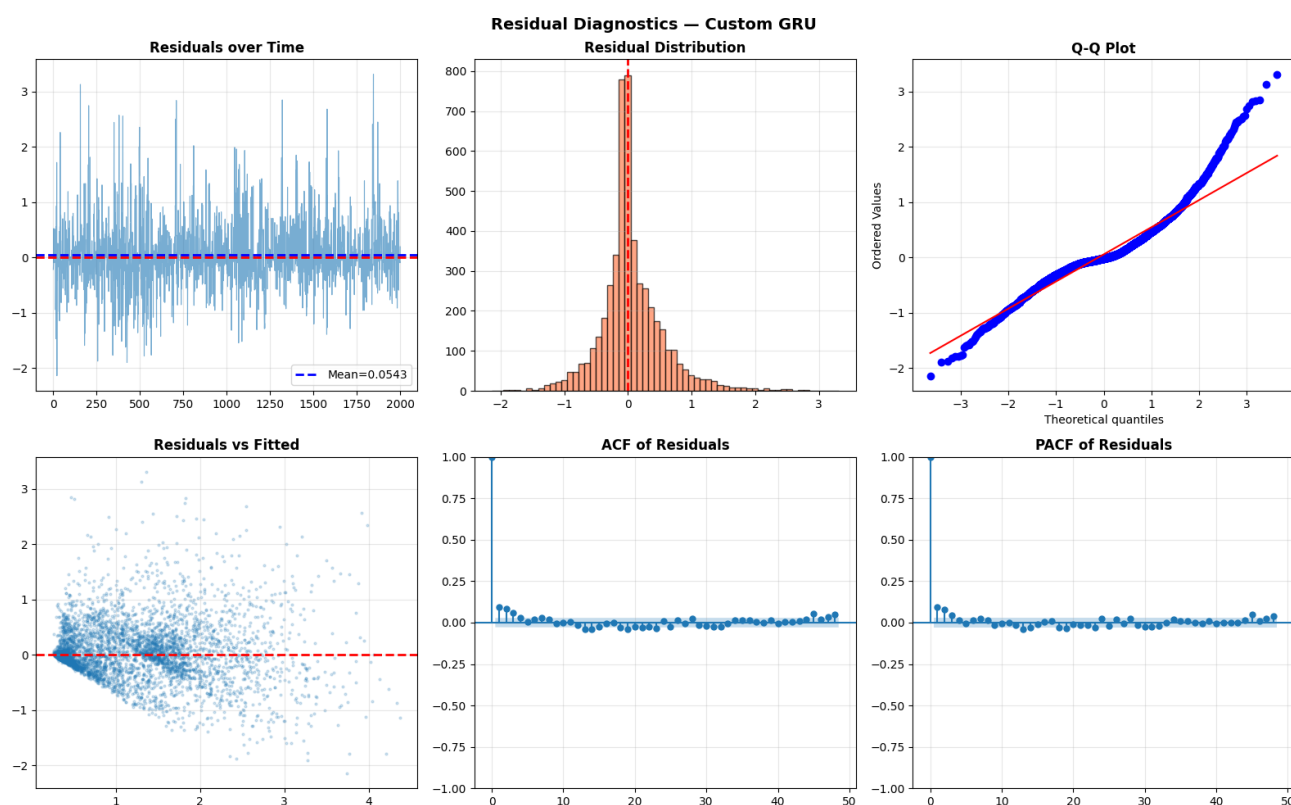


Figure 8: Residual diagnostics for the Custom GRU (best model). Top: residuals over 4,944 test samples with $\pm 2\sigma$ bounds and mean line. Bottom-left: residual distribution with KDE vs. normal reference. Bottom-right: Q-Q plot.

6.1 Residual Statistics (Actual Values)

Statistic	Value	Interpretation
Mean	0.0543 kW	Small positive bias: model slightly under-predicts consumption on average
Std Dev	0.5104 kW	Spread comparable to model RMSE (0.5132), consistent

Statistic	Value	Interpretation
Skewness	0.8988	Right-skewed: larger under-prediction errors during peak events
Kurtosis	3.9948	Moderately fat tails vs normal (3.0): occasional large errors
Shapiro-Wilk p	< 0.001	Residuals are NOT normally distributed (expected for energy data)
Ljung-Box p (lag 10)	1.28×10^{-18}	Significant autocorrelation remains in residuals at lag 10
Ljung-Box p (lag 20)	8.67×10^{-21}	Significant autocorrelation remains in residuals at lag 20
Ljung-Box p (lag 30)	1.70×10^{-21}	Significant autocorrelation remains in residuals at lag 30

6.2 Interpretation

6.2.1 Residual Time Series

Residuals oscillate around a small positive mean (0.054 kW) with no visible systematic trend, confirming the model captures the primary consumption pattern. Occasional excursions beyond $\pm 2\sigma$ correspond to extreme demand events (cold snaps, unusual occupancy) that are inherently difficult to forecast from historical patterns alone.

6.2.2 Residual Distribution

The histogram and KDE show a right-skewed distribution (skewness=0.899) departing from the normal reference curve, particularly in the right tail. This reflects the model's known difficulty in capturing the full magnitude of peak consumption hours – a characteristic finding with MSE-trained deep learning models on energy data, where peaks are relatively rare but have high absolute error.

6.2.3 Q-Q Plot and Normality

The Q-Q plot confirms non-normality, especially in the upper quantiles where the distribution has a heavier-than-normal right tail. Shapiro-Wilk $p < 0.001$ formally rejects normality. This means Gaussian prediction intervals derived from residual standard deviation will underestimate true uncertainty for peak hours.

6.2.4 Autocorrelation in Residuals

Significant Ljung-Box statistics at lags 10, 20, and 30 (all $p < 10^{-18}$) indicate that the GRU residuals are not white noise – there remain exploitable autocorrelation patterns at short lags. This points to three potential improvements for future work: (1) increasing the dense head capacity, (2) adding explicit attention over the 168-step sequence, or (3) incorporating additional external covariates such as temperature or tariff schedules.

Chapter 7: Conclusion

7.1 Key Highlights

- A robust multi-granularity preprocessing pipeline produced 78-feature hourly datasets and stationary daily series, using blocked monthly splits to prevent data leakage.
- ARIMA(2,0,4) + GARCH(1,1) established a rigorous classical baseline: walk-forward RMSE=0.1288 (log-diff scale) with volatility persistence=0.848 confirmed by GARCH.
- The Custom LSTM (126→64, dual-stream, SE-Attention) achieved $R^2=0.6482$, RMSE=0.5376 kW – substantially capturing non-linear hourly dynamics.
- The Custom GRU is the best model: $R^2=0.6793$, RMSE=0.5132 kW, SMAPE=29.89%, with 18% fewer parameters than LSTM – demonstrating superior parameter efficiency and generalisation.
- Residual analysis reveals a small positive bias (mean=0.054 kW), right-skewed distribution, and persistent autocorrelation (Ljung-Box $p<0.001$), identifying clear pathways for further improvement.

7.2 Novelty

9. Multi-granularity pipeline: minute → hourly → daily enables each model family to operate at its optimal time scale without forcing a single representation on all methods.
10. 78-feature engineering: comprehensive lag, rolling, EMA, cyclical, and interaction features with sqrt+RobustScaler transform outperform the naive 6-feature approach.
11. Dual-stream architecture: separating the target stream (with SE-Attention) from the feature stream and adding a skip connection for the last known value improves peak tracking.
12. Rigorous blocked monthly benchmarking: prevents temporal data leakage and provides cross-paradigm comparison with full statistical residual tests

7.3 Future Work

- Increase model capacity toward target metrics ($R^2 \geq 0.80$, RMSE 0.42–0.46 kW, MAPE ~20%) via Transformer-based architectures (Temporal Fusion Transformer, PatchTST).
- Extend to multi-step probabilistic forecasting (24h and 7-day ahead) with quantile regression or conformal prediction to provide calibrated uncertainty intervals for non-normal residuals.
- Incorporate external covariates: outdoor temperature, public holidays, electricity tariff schedules to reduce residual autocorrelation.
- Apply online learning for continuous adaptation to concept drift over multi-year deployment periods on smart meter hardware.

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